

Introduction to Numerical Weather Prediction



Chapter 4: Parameterization Of Physical Processes

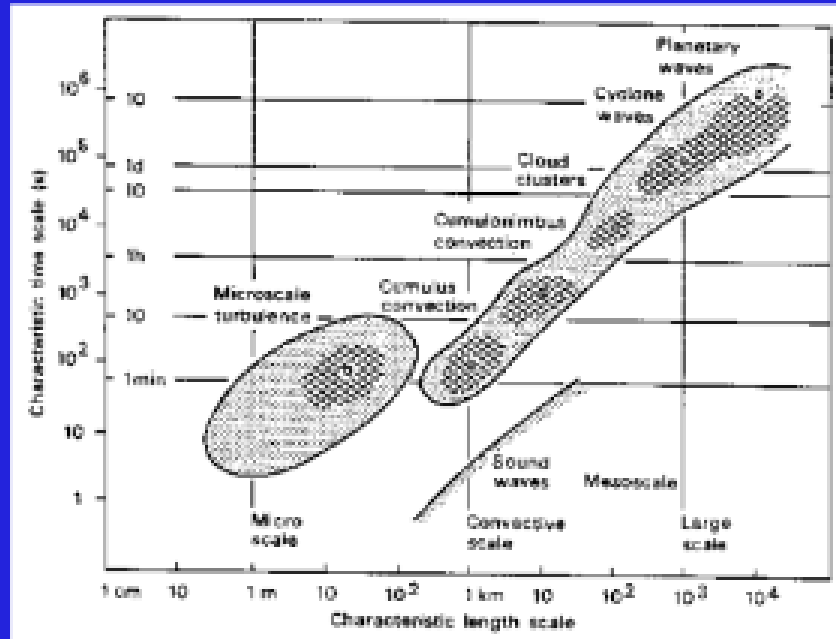


Objective

- *To Define Physical Parameterizations*
- *To Know The Necessity of Parameterizations in NWP*
- *To Identify the Processes to be parameterize*
- *To Know how do Forecasts Go Bad*

Parameterization Of Physical Processes

What is physical parameterization?



Characteristic scales of
atmospheric processes

- **Atmospheric motions have different scales.**
- **Climate model resolutions:**
Regional: 50 km
Global: 100~200 km
- **Sub-grid scale processes:**
Atmospheric processes with scales can not be explicitly resolved by models.
- **Physical parameterization:**
To represent the effect of sub-grid processes by using resolvable scale fields.

Physical Parameterizations

- Small scale processes are not resolved by large scale models, because they are sub-grid.
- The procedure of expressing the effect of sub-grid process is called parameterization.
- Each important physical process that cannot be directly predicted requires a parameterization scheme based on reasonable physical or statistical representations.

Physical Parameterizations

- Used to approximate the bulk effects of physical processes too small, brief, complex or poorly understood to be explicitly represented by the governing equations and/or numerical methods.
- because they cannot be explicitly predicted in full de-tail in model forecast equations.
- Much of the weather occurs at scales smaller than those resolved by the weather forecast model. Model must treat, or “parameterize” the effects of the sub-grid scale on the resolved scale.

Why do we need to parameterize?

- NWP models can not resolve weather feature and/processes that occurs within a single model grid box.
- However, *the model must include or take in to account the effect of this processes because they have a significant role in predicting weather.*
- Taking in to account these effect with out actually simulating them is called a Parameterization.

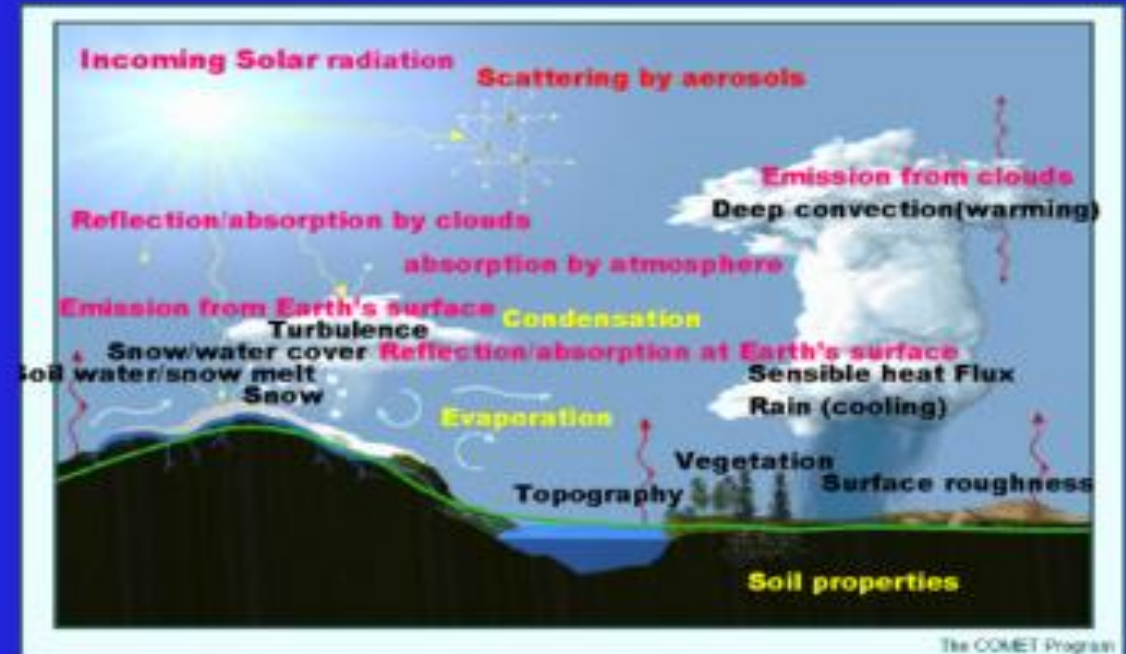
Why do we need to parameterize?

- Therefore parameterization is necessary for the following reasons
 - ❖ Computers are not yet powerful enough to treat many physical processes explicitly *because they are either too small or complex to be resolved.*
 - ❖ Many other physical processes can not be explicitly modeled *because they are not sufficiently understood to be represented in equation format or there are no appropriate data.*
 - ❖ There are many complex processes that needs to be parameterized....

What should be parameterized ?

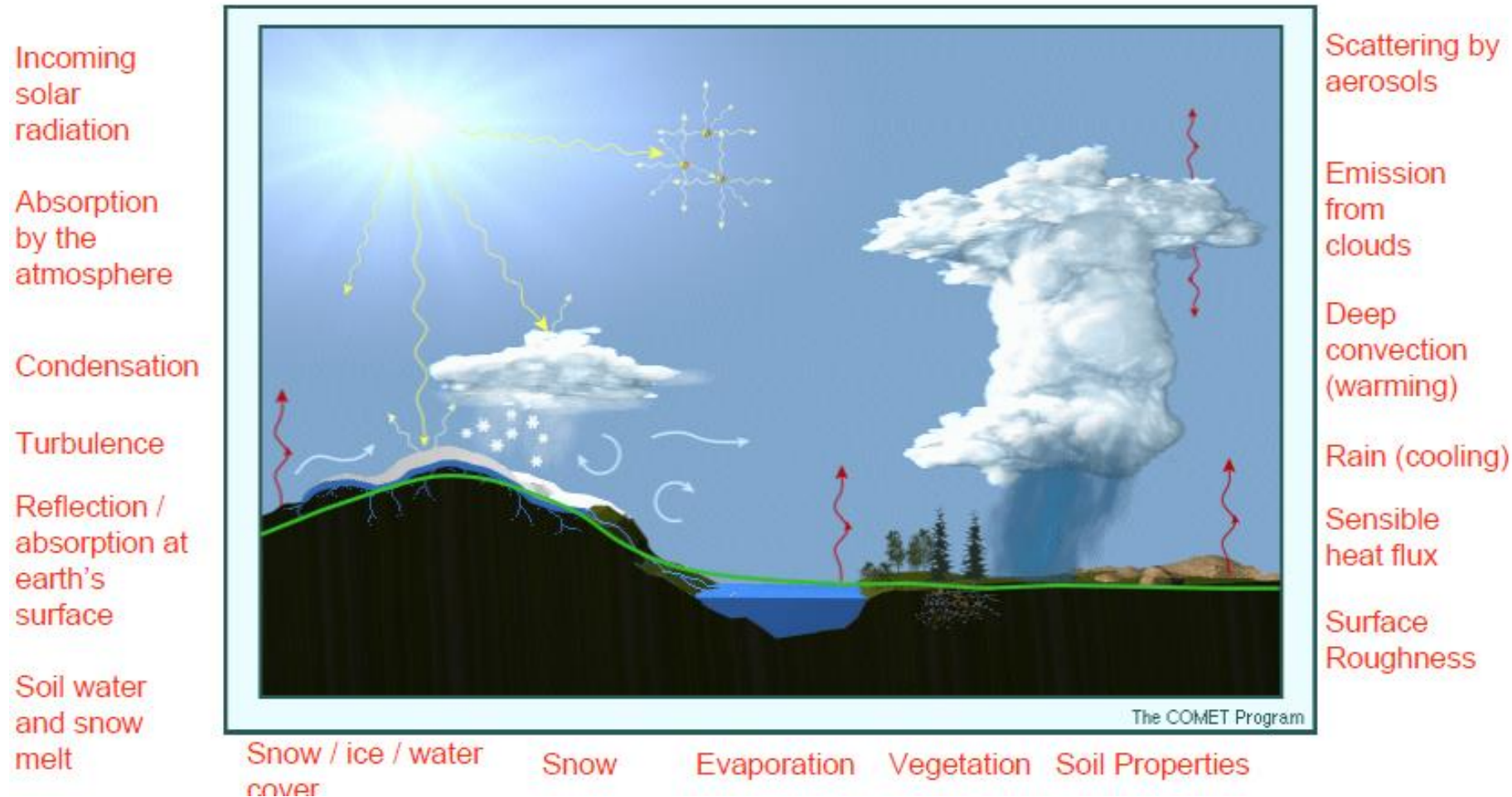
Model Physics include:

- Radiation transfer.
- Surface processes.
- Vertical turbulent processes.
- Clouds and large-scale condensation.
- Cumulus convection.
- Gravity wave drag.



16 major physical processes in climate system. (from <http://www.meted.ucar.edu/nwp/pcu1/ic4/frameset.htm>)

- The processes shown below are often parameterized because:
 - they cannot be explicitly predicted in full detail by model forecast equations in spite of the model resolution
 - their effects are important to the simulation

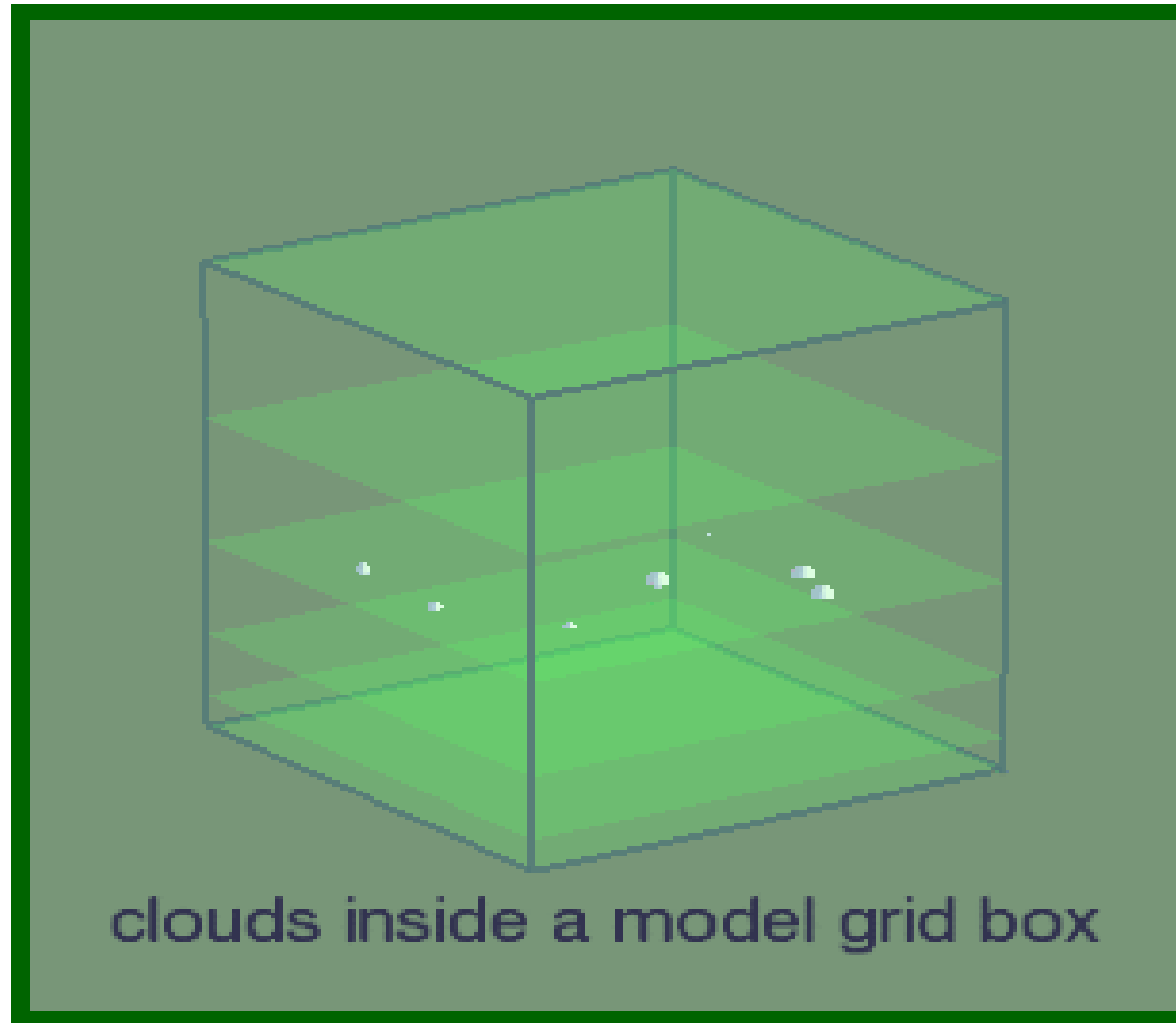




- 1) Incoming Solar Radiation
- 2) Scattering by Aerosols and Molecules
- 3) Absorption by the Atmosphere
- 4) Reflection/Absorption by Clouds
- 5) Emission of Longwave Radiation from Earth's Surface
- 6) Condensation
- 7) Turbulence
- 8) Reflection/Absorption at Earth's Surface
- 9) Snow
- 10) Soil Water/Snow Melt
- 11) Snow/Ice/Water Cover

- 12) Topography
- 13) Evaporation
- 14) Vegetation
- 15) Soil Properties
- 16) Rain (Cooling)
- 17) Surface Roughness
- 18) Sensible Heat Flux
- 19) Deep Convection (Warming)
- 20) Emission of Longwave Radiation from Clouds

- *Schematic illustration of small scale clouds and their development within a grid box of a NWP model*



- The animation above illustrates this kind of situation for an example where a number of small scale clouds exist within a model grid box, which is subdivided vertically by a number of layers.
- Within this grid structure, clouds can develop and while they exist interact with other small scale processes like turbulence or radiation.
- They might also lead to the generation of precipitation that would then fall only on a part of the surface of the earth below the model grid box.
- Apart from their impact in form of precipitation or a modification of the solar radiation available at the surface, which are key forecast products of a NWP model, there exists also an interaction between the sub-grid scale and grid scale processes, which implies that the small scale processes influence also the entire numerical weather forecast.

- Procedures to describe those processes which can not be explicitly resolved by the model grid are known under the term ,physical parameterization‘. Which processes fall into this category depends directly on the spatio-temporal grid structure employed by the forecast model.

Radiation Parameterization

Why is Radiation Important?

- The sun is the primary source of energy for the Earth.
- Most of this energy comes in the form of radiation energy, often referred to as insolation.
- Insolation, or more specifically spatial and temporal variability therefor, drives atmospheric circulations!
 - Hadley circulation; mid-latitude westerlies; tropics; etc.
 - Moist convection; coastal circulations; inversions; etc.

Why is Radiation Important?

- Insolation is a function of many different things...
 - Latitude and the seasonal cycle
 - Angle at which insolation intersects the Earth
 - *Attenuation by the Earth's atmosphere*
- A model must be able to accurately simulate how insolation interacts with land, oceans, vegetation, clouds, air molecules, and aerosols.

Why is Radiation Important?

- These interactions are typically parameterized by numerical models.
- Important radiative transfer processes (scattering, absorption, etc.) occur on the molecular level.
- Radiative transfer is complex...
 - Dependency upon a broad spectrum of wavelengths
 - Dependency upon numerous atmospheric constituents
 - Oftentimes considered over many distinct vertical layers, involving numerous calculations per layer

Fundamentals of Radiation

- There are four important processes to consider when attempting to describe radiative transfer...
 - **Absorption**: radiation energy absorbed by the surface and/or one or more atmospheric constituents
 - **Scattering**: radiation energy reflected by the surface and/or one or more atmospheric constituents
 - **Emission**: radiation energy emitted by the surface and/or by one or more atmospheric constituents
 - **Transmission**: radiation energy that is transmitted from space to the Earth's surface (or vice versa)

Fundamentals of Radiation

- Energy transfer in the atmosphere involves radiation in two distinct bands of wavelength...
 - **Shortwave**: radiation emitted by the sun
 - **Longwave**: radiation emitted by the Earth's surface and atmosphere

Fundamentals of Radiation

- Longwave radiation released by the Earth's surface is almost completely absorbed by the atmosphere.
 - H₂O, over a wide band centered at 6.3 μm
 - CO₂, over a band centered at 15 μm, or near the peak of the longwave radiation emission spectrum
 - Trace gases (O₃, CH₄, N₂O) at specific wavelengths
- What is absorbed at some level is subsequently re-emitted, with half directed upward and half directed downward.

Fundamentals of Radiation

- Most longwave radiation emitted by the Earth's surface is absorbed in the troposphere.
- Only a small amount of longwave radiation reaches beyond the tropopause.

Fundamentals of Radiation

- A couple of relevant definitions...
 - **Albedo**: Fractional amount of incoming solar radiation that is reflected and/or scattered back to space by the Earth's surface, atmosphere, and clouds.
 - **Emissivity**: Fractional amount of radiation (per unit area per unit time) emitted by some emitter such as the Sun or the Earth's surface.
- Larger values reflect greater reflection and greater emission, respectively.

Radiation Parameterization

- A radiation parameterization must be able to address variability in the context of radiative transfer.
- It must be able to accurately describe how radiation is absorbed, emitted, scattered, and transmitted within the entire model atmosphere.
- Most radiation parameterizations will do this for longwave ***or*** shortwave radiation in terms of a **radiation flux**.

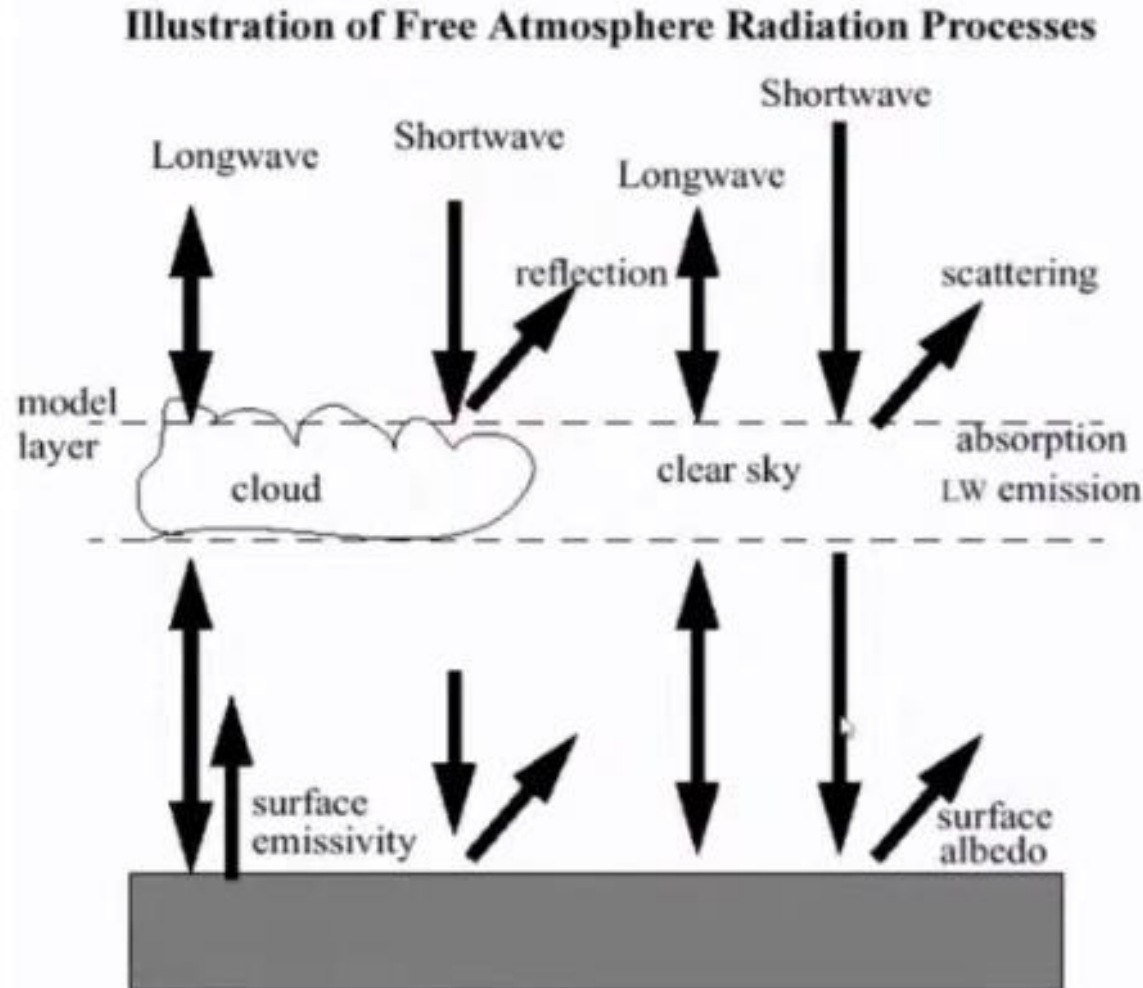
Radiation Parameterization

- The challenge in radiation parameterization is a classic one: accuracy versus efficiency.
 - A radiation parameterization is generally not called each model time step because it is computationally intensive.
 - Accuracy is also strongly influenced by the coupled nature of atmospheric processes to radiation effects.

Radiation Parameterization

- Radiation parameterization requires accurate knowledge of both atmospheric state and non-routinely observed molecular constituent fields.
 - Atmospheric state: temperature (Planck's Law, Wien's Displacement), water vapor concentration (H_2O), etc.
 - Molecular constituents: particularly natural and anthropogenic aerosols (CO_2 , O_3 , CH_4 etc.)
- This poses particular challenges to climate models...
 - Accurate representation of aerosol radiative effects
 - Accurate knowledge of future aerosol concentrations

Radiation Transfer

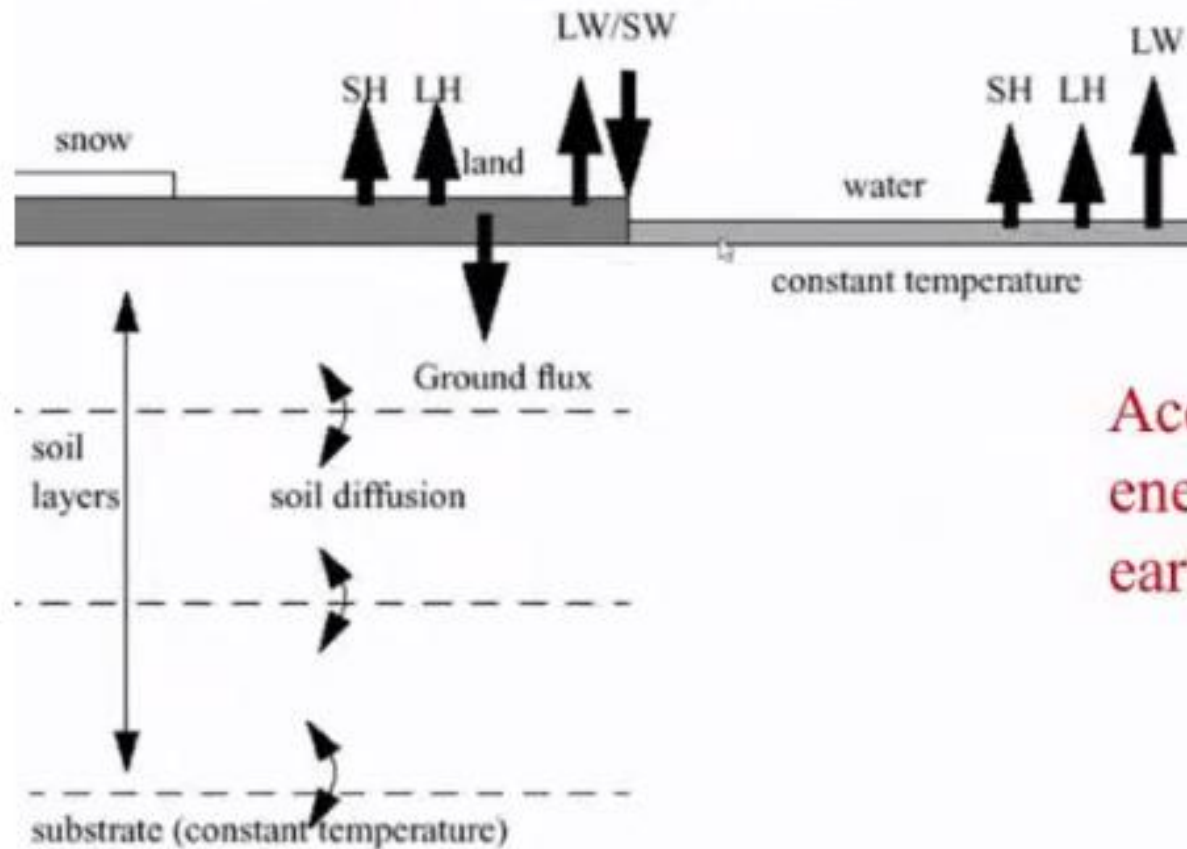


Accounts for the effect of radiative processes in the atmosphere

Surface Fluxes

Surface Process

Illustration of Surface Processes

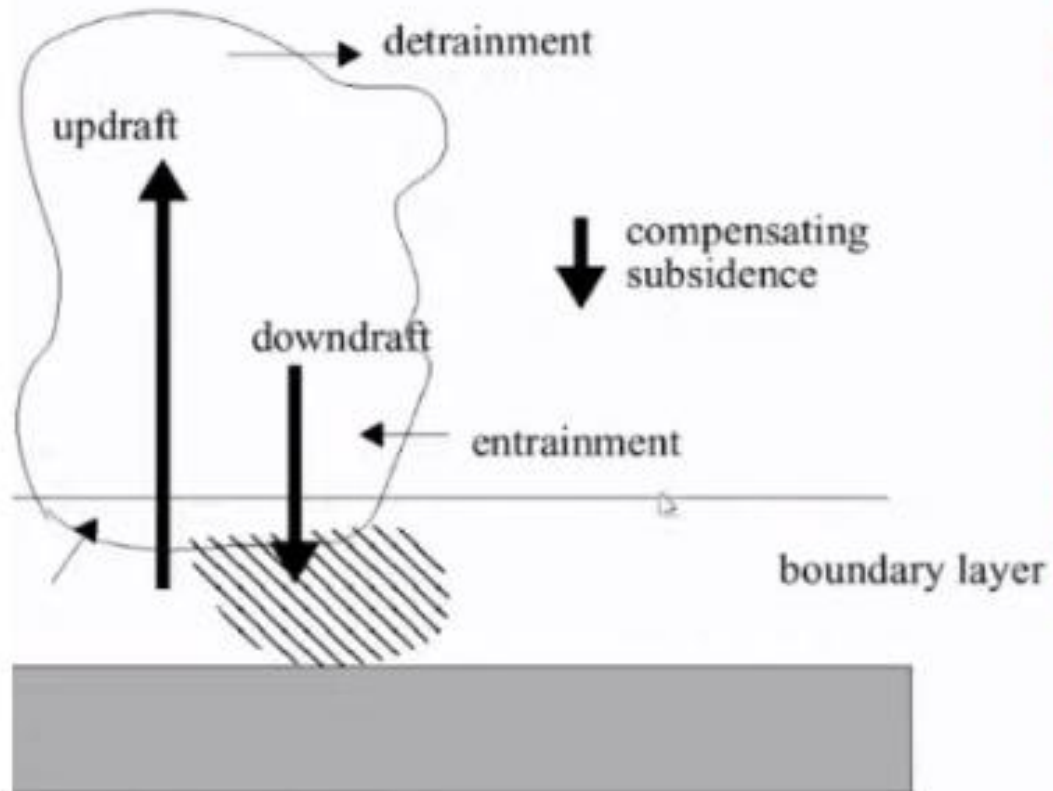


Accounts for the exchange of energy and water at the earth's surface

Cumulus convection

Cumulus Processes

Illustration of Cumulus Processes



Accounts for the collective influence of small scale convective processes on large scale model variables

Influence vertical stability and large scale flow pattern by redistributing heat, moisture and momentum

How do we do parameterization in numerical models?

- **Ignore some processes (in simple models).**
- **Simplifications of complex processes based on some assumptions.**
- **Statistical/empirical relationships and approximations based on observations.**
- **Nested models and super-parameterization:**
Embed a cloud model as a parameterization into climate models.

The problems in parameterization

- **The current parameterization schemes are too simple to describe the nature of the processes.**
- **Our knowledge about physical processes and feedback mechanism limits the improvement of parameterization.**
- **Superparameterization seems to be a better way to represent of physical processes comparing with conventional parameterization.**
- It is only used in cloud processes (CRM).
- Computational costs are very expensive, about 100 ~ 1000 times more than the conventional parameterization.

Summary



- **Parameterization is a method to represent the effects of physical processes which are too small or too complex or poorly understood.**
- **The importance of parameterization for weather and climate prediction has been well recognized and a lot of works have been done to improve physical parameterization. But, parameterization has not been a mature subject till now.**
- **The best way to improve parameterization is to understand the physical processes better by observations and high resolution simulations .**

....END

Thanks !